Name: _____ ID Number: _____

THE UNIVERSITY OF THE SOUTH PACIFIC

School of Computing, Information and Mathematical Sciences, Faculty of Science, Technology and Environment

IS222: Database Management Systems *(Online)*

FINAL EXAMINATION – Semester 2, 2013

Time Allowed 3 hours plus 10 minutes reading

89 marks (40% of final grade)

INSTRUCTIONS

1. This exam has two sections with **23 questions**:
   a. Section A : 20 marks
   b. Section B : 69 marks
2. **Answer all questions** in **sections A and B** in this **same booklet** ***(17 pages)***.
3. You may use calculators.
4. This exam is worth 40% of your overall mark. Minimum pass mark (40%) of this exam which is 16%/40%. Also, you need a minimum 60% attendance of the labs to attend the exam.
5. For **Section A**
   a. Select a multiple-choice answer by putting a circle around that letter.
   b. Correct mistakes by putting an "X" through the circle or erasing it completely.
   c. If you put an "X" through a circle, it will be considered unmarked.
   d. Ambiguous marks will be counted as incorrect.
6. For **Section B**, place your answers in the booklet in the spaces provided for each question.

You **cannot** take ***this*** examination booklet with you when you complete the examination. *This is not an open book exam*

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Section A – Multiple Choice *(20 mks)*

Place your Multiple Choice answer ***here only*** *by circling the correct answer. Cross out any incorrectly circled answer with an X, to replace it. The questions are on the next page.*

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D
11	A	B	C	D
12	A	B	C	D
13	A	B	C	D
14	A	B	C	D
15	A	B	C	D
16	A	B	C	D
17	A	B	C	D
18	A	B	C	D
19	A	B	C	D
20	A	B	C	D

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1. Which of the following is true about Business rules (BR)?

   a. A BR always provides a detailed description of the end user
   b. A BR is concise and ambiguous
   c. A BR does not depict the cardinality in a relationship
   d. A BR is concise and unambiguous

2. Dependencies can be identified with the help of the dependency

   a. ERD
   b. schema
   c. form
   d. diagram

3. The primary key is selected from the:

   a. composite keys
   b. foreign keys
   c. determinants
   d. candidate keys

4. A table is in _____ and has no multivalued dependencies:

   a. second normal form when it is in 1NF
   b. domain/key normal form when it is in 2NF
   c. fourth normal form when it is in 3NF
   d. third normal form when it is in BCNF

5. Row is synonymous with the term:

   a. record
   b. attribute
   c. column
   d. field

6. Which of the following indicates the minimum number of entities that can be involved in a relationship?

   a. Minimum cardinality
   b. ERD
   c. Maximum cardinality
   d. Greater Entity Count (GEC)

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7. A _____ key is defined as a key that is used strictly for data retrieval purposes.

   a. lookup
   b. foreign
   c. candidate
   d. secondary

8. A view is which of the following?

   a. A virtual table that cannot be accessed via SQL commands
   b. A virtual table that can be accessed via SQL commands
   c. A base table that can be accessed via SQL commands
   d. A base table that cannot be accessed via SQL commands

9. SQL data definition commands make up a(n) _____ .

   a. HTML
   b. DDL
   c. DML
   d. XML

10. Consider the relationship description: A student can take many courses and a course can have many students. The best way of representing this relationship between courses and students from the choices below is:

    a. A one-to-one relationship between course and student
    b. There is no relationship possible
    c. To have a one-to-many relationship between the entities via a composite entity
    d. A direct many-to-many relationship between course and student

11. Which of the following best relates to data?

    a. It can never have anomalies
    b. It consists of raw facts
    c. It's a programming language
    d. It cannot be stored

12. Which of the following is the correct ERD for the business rule stated below?

    A project is managed by exactly one employee, and each employee can manage one or more projects.

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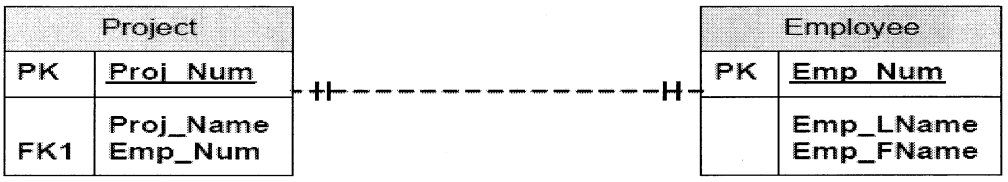
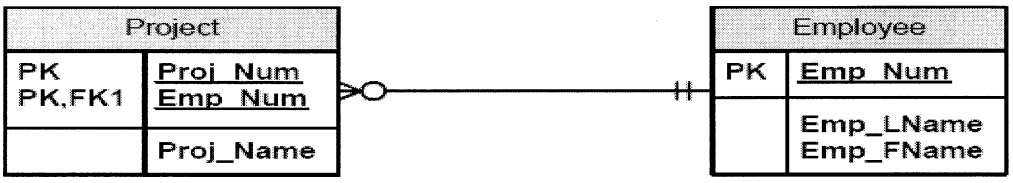
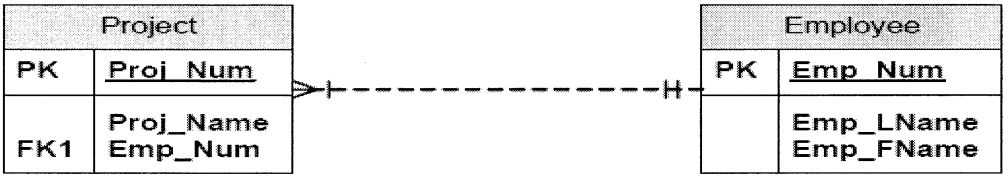
A.



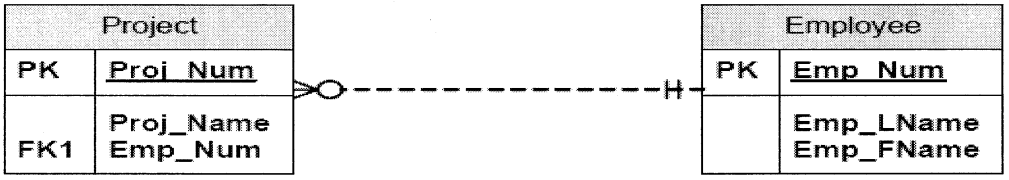
B.



C.



D.



13. Which of the following statements concerning the primary key (PK) is true?

    a. A PK must be null
    b. A PK can be duplicated in the same table
    c. A PK is not required in a table
    d. A PK cannot be null

14. When looking at transaction management in databases, every individual transaction must display some key properties which are sometimes referred to as the:

    a. concurrency test
    b. ACID test
    c. granularity test
    d. deadlock resilience

15. In transaction management, which of the following is true regarding Two-phase locking?

    a. It prevents deadlocks
    b. It consists of a growing phase in which a transaction releases all locks and cannot obtain any new lock

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c. No lock operation can precede a lock operation in the same transaction
d. It may not be able to guarantee serializability at all times

16. ANSI has defined standards that govern SQL database transactions. The SQL statement(s) that provide transaction support include *(choose the best answer)*:

a. ROLLBACK
b. COMMIT, ASSIGN and ROLLBACK
c. UNLOCK
d. None of the Above

17. A(n) ____ join links tables by selecting only the rows with common values in their common attribute(s).

a. equal
b. unique
c. foreign
d. natural

18. Which of the following relate to DML?

a. Select
b. Insert
c. Update
d. All of the above

19. A _____ attribute can be further subdivided to yield additional attributes.

a. composite
b. simple
c. single-valued
d. multivalued

20. A _____ relationship exists when an association is maintained within a single entity.

a. unary
b. binary
c. ternary
d. weak

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Section B *(69 mks)*

Answer all questions in the spaces provided.

Question 21 – Entity Relationship Modelling *(34.5 mks)*

Scenario:

You have been hired by a new college called Tiny ERD to design a fundamental and centralized relational database in MySql. As part of the systems requirements process, the following information about how the college records its data has been uncovered:

The college has many students and the details to be recorded include the following: student's first name, last name, residential address and email only. The college also has a school and departmental structure. Each school has a specific name which must be recorded. There are also several departments and for each the database must record it's particular name.

The college also has employed several professors, each of whom has a teaching specialty, rank (eg. Senior, Assistant), first name, last name and email. These must all be recorded. There are also many buildings in the college. Each building has a specific name and location that must be recorded.

Students attend classes and there are important details about each class that needs to be recorded. These include the class day (eg. Monday), class time (eg. 11-1pm) and a section (eg. Section 3, Section 2 etc). These classes are part of specific courses. The course info that must be recorded includes: it's title, description and if it can be cross credited.

Apart from the details above that need to be recorded, there are other details to be noted about how Tiny College runs. Firstly, each school in Tiny College has several departments whilst each department belongs to one major school. The departments play a key role in offering courses in the college. Each department may or may not offer a set of courses. Every course that runs in the college is associated with a specific department Additionally, a course may or may not have many

classes scheduled for it by the administrators. A class that runs for the college is always associated with only one particular course. The venue for each class is one of the buildings in the college. A particular building may host many classes. Sometimes a building may have no classes at all. Each particular class always runs in a specific building like the N111 building. This is largely since there is no need for video broadcasting and student numbers are not that large.

Additionally, a student may take several classes or not take a class at all. Each class generally has many students in it, but early on the classes may be scheduled to exist even though no students have enrolled in it. This could happen during the enrollment period early on in the semester.

Finally, since the college has received additional funds from the government, each school has at least one professor. Each professor is, however, designated to a specific school

1. List any business rules that you have used to design the database before drawing your ERD. **These should be concise. (12 marks)**.

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2. Draw out your **Crow's Foot Entity Relationship Diagram (ERD)**. Be sure to **specify** the **entity names, attributes**, **primary** and **foreign keys, cardinalities** and **relationship strengths. (22.5 marks)**.

Question 22 – Normalization (12 *mks)*

Table 1 below shows a **Project related database** with **some sample records**. You will use this data and decompose it as part of the normalization exercise questions below.

Table 1

PROJ_NUM	PROJ_NAME	EMP_NUM	EMP_NAME	JOB_CLASS	CHG_HOUR	HOURS
15	Evergreen	103	June E. Arbough	Elect. Engineer	84.50	23.8
15	Evergreen	101	John G. News	Database Designer	105.00	19.4
15	Evergreen	105	Alice K. Johnson *	Database Designer	105.00	35.7
15	Evergreen	106	William Smithfield	Programmer	35.75	12.6
15	Evergreen	102	David H. Senior	Systems Analyst	96.75	23.8
18	Amber Wave	114	Annelise Jones	Applications Designer	48.10	24.6
18	Amber Wave	118	James J. Frommer	General Support	18.36	45.3
18	Amber Wave	104	Anne K. Ramoras *	Systems Analyst	96.75	32.4
18	Amber Wave	112	Darlene M. Smithson	DSS Analyst	45.95	44.0
22	Rolling Tide	105	Alice K. Johnson	Database Designer	105.00	64.7
22	Rolling Tide	104	Anne K. Ramoras	Systems Analyst	96.75	48.4
22	Rolling Tide	113	Delbert K. Joenbrood *	Applications Designer	48.10	23.6
22	Rolling Tide	111	Geoff B. Wabash	Clerical Support	26.87	22.0
22	Rolling Tide	106	William Smithfield	Programmer	35.75	12.8
25	Starflight	107	Maria D. Alonzo	Programmer	35.75	24.6
25	Starflight	115	Travis B. Bawangi	Systems Analyst	96.75	45.8
25	Starflight	101	John G. News *	Database Designer	105.00	56.3
25	Starflight	114	Annelise Jones	Applications Designer	48.10	33.1
25	Starflight	108	Ralph B. Washington	Systems Analyst	96.75	23.6
25	Starflight	118	James J. Frommer	General Support	18.36	30.5
25	Starflight	112	Darlene M. Smithson	DSS Analyst	45.95	41.4

Hint: When trying to locate the determinants, look for **each field** that **influences** another **field consistently** in **relation** to the **data presented**. This table will use a **composite primary key**, so a **combination of fields** may be **required** to **uniquely identify each record**.

1. Use the following diagram to create the **1NF dependency diagram** for the above table. Clearly label the **Partial** and **Transitive dependencies**. **Show all dependencies in the 1NF**. **(4 marks)**

PROJ_NUM	PROJ_NAME	EMP_NUM	EMP_NAME	JOB_CLASS	CHG_HOUR	HOURS

2. Using the initial dependency diagram you have given in Question 1 above. **Convert your table structure to 2NF**, and draw the new dependency diagram**. (4 mks)**

3. Using the table structures you created in Question 2 **and convert your table structure** to **3NF.** Draw the new **3NF dependency diagram**. **(4 marks)**

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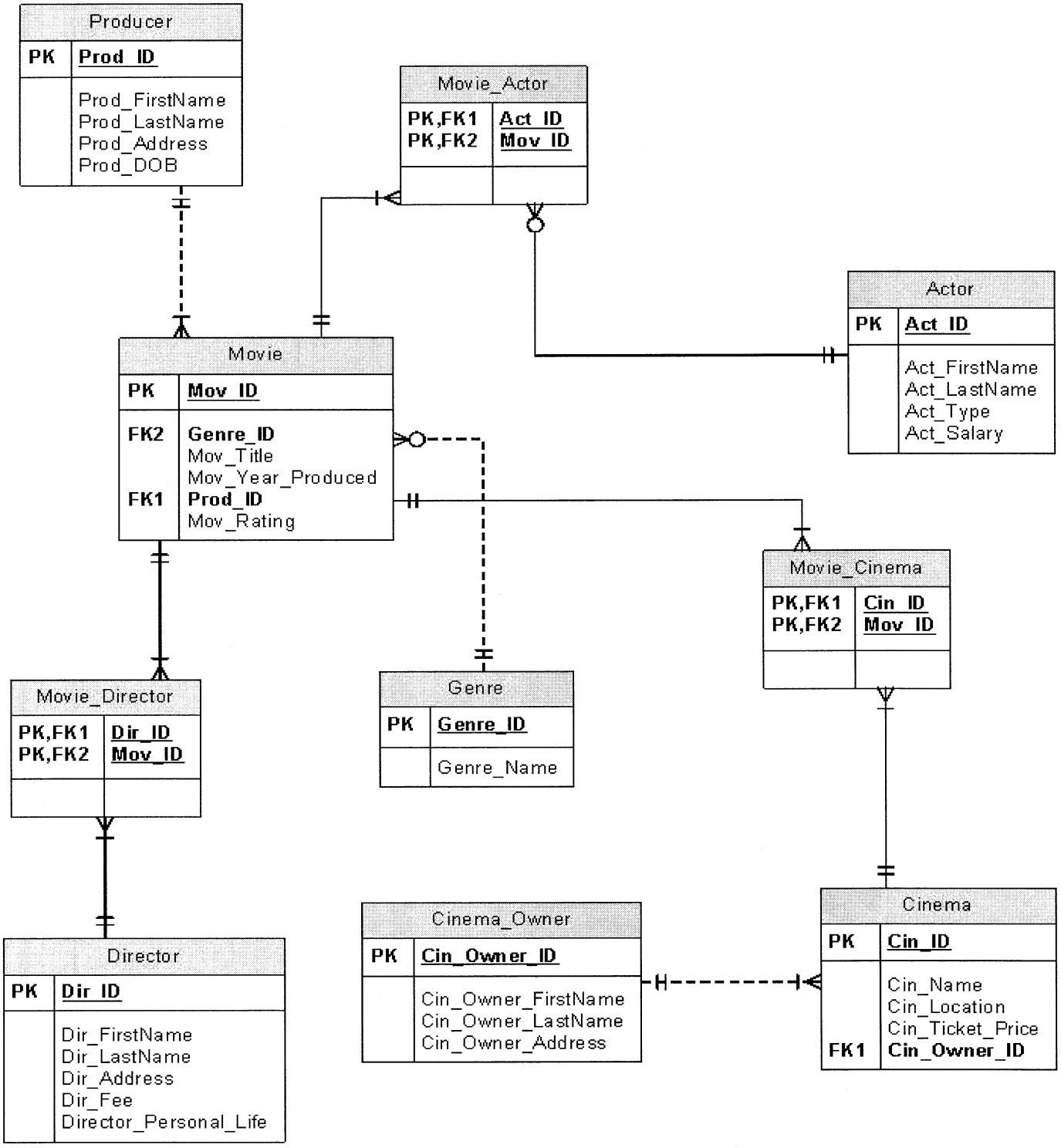
Question 23 – Structured Query Language (SQL) – *22.5 mks*

Refer to the ERD below to answer the following **SQL related questions**. All the queries and commands you write will be for a **MySql database**. Assume that a **MySql database** called **production** is created and you will write **all queries (DDL/DML) related** to that **MySql database**. Answer the questions that follow.



a) Write the code to login to the MySql account from the command line using the mysql command. Use the username, root and password, test. **(2mks).**

b) After the command in (a) above is issued and you are connected to the MySql server with the root account, issue the command to connect to the database called production on the console **(1mk)**.

c) Write a query to list all the details of all the actors in descending order of their Last Name (Act_LastName). Do not list all the attributes individually in your query, use an alternative method. **(2mks).**

d) Write a query to list only the salary of all actors with the first name of Tom. **(2mks).**

e) Write a query to update the cinema record with the id, C3, with the new location (Miami, Florida) and ticket price (29.99). **(2.5mks).**

f) Write a query to delete the genre record with the id, G2. **(1mks).**

g) **For each producer,** list each producer's first and last name and the total number of movies aliased as Num_Movies produced **by that** producer ***(4mks).***

h) **For each movie,** list its movie name (aliased as Movie_Name) and director's full name *(aliased as Director_FullName)*. See a sample output below: ***(3mks).***

Movie_Name	Director_FullName
1408	Quentin Tarantino
What Woman Want	Frank Darabont
National Treasure	Quentin Tarantino
Let Me In	Gore Verbinski

i) Write a query to create the Movie_Director table. Assume that the fields Dir_ID and Mov_ID are Strings that are no more than 10 characters in length. Use the Innodb engine. ***(5mks).***

End of Section B
End of EXAM